

Radiative transitions of B_{c1} states in hard-photon QCD sum rules

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Abstract

We study the radiative transitions $B_{c1}(6743) \rightarrow B_c \gamma$, $B_{c1}(6743) \rightarrow B_c^* \gamma$, $B_{c1}(6750) \rightarrow B_c \gamma$, and $B_{c1}(6750) \rightarrow B_c^* \gamma$ in a heavy-heavy three-point QCD sum-rule framework. Since both valence quarks are heavy, the leading radiative contribution is hard photon emission from the charm and anti-bottom lines rather than a photon-distribution-amplitude term. Using physical-current residues from the two-point B_c -mixing sum rules, we find

$$\Gamma[B_{c1}(6743) \rightarrow B_c \gamma] = 0.223_{-0.061}^{+0.084} \text{ keV}, \quad \Gamma[B_{c1}(6750) \rightarrow B_c \gamma] = 8.01_{-1.41}^{+1.89} \text{ keV}.$$

For the vector final state, using the standard vector-current normalization $f_{B_c^*} = 0.435_{-0.035}^{+0.032} \text{ GeV}$, we obtain the leading hard-photon baselines

$$\Gamma[B_{c1}(6743) \rightarrow B_c^* \gamma] = 74.3_{-9.0}^{+11.2} \text{ keV}, \quad \Gamma[B_{c1}(6750) \rightarrow B_c^* \gamma] = 360_{-48}^{+52} \text{ keV}.$$

We also give a conservative dimension-four gluon-condensate screening uncertainty and show that the vector-channel contact terms have no ordinary two-channel double-pole support.

I. INTRODUCTION

The spectroscopy of the B_c family is a particularly clean laboratory for heavy-heavy dynamics. Unlike ordinary heavy-light mesons, the B_c mesons contain two different heavy flavors, and their electromagnetic transitions are sensitive to the relative photon couplings of the charm and anti-bottom quark. Radiative transitions of orbitally excited B_c states therefore provide a useful complement to mass spectroscopy and hadronic transitions.

The observation of structures in the $B_c^+ \gamma$ spectrum by LHCb [1, 2] motivates a direct QCD sum-rule calculation of the corresponding radiative widths. The experimental analysis observes peaks in the visible $B_c^+ \gamma$ invariant mass, but does not measure the individual partial radiative widths. Theoretical predictions for $B_{c1} \rightarrow B_c^{(*)} \gamma$ have been obtained in several potential model and relativistic quark-model approaches [3–11]. The spread among predictions is sizeable, especially for channels in which the spin components of the axial state interfere.

In the strange heavy systems, light-cone QCD sum rules receive important photon-distribution-amplitude contributions. The B_c case is different: there is no light valence quark, and the leading radiative term is the short-distance photon emission from heavy

propagators. In this paper we build a hard-photon three-point QCD sum rule for the transitions $B_{c1} \rightarrow B_c \gamma$ and $B_{c1} \rightarrow B_c^* \gamma$. The axial-vector currents and residues are matched to the same physical-current basis used in the B_c -mixing two-point analysis. We also give a detailed audit of the dimension-four gluon-condensate sector, since in a heavy-heavy system it is the first natural nonperturbative correction to the perturbative triangle.

The paper is organized as follows. In Sec. II we introduce the interpolating currents, the physical mixing convention, and the radiative form factors. Section III summarizes the hard-photon sum-rule construction and the condensate audit. Numerical inputs, stability checks, and comparison with experiment and other theory are given in Sec. IV. Section V summarizes the results.

II. THEORETICAL FRAMEWORK

We use two independent axial-vector basis currents,

$$J_\mu^A = \bar{c} \gamma_\mu \gamma_5 b, \quad (1)$$

$$J_\mu^B = \frac{i}{m_b + m_c} \bar{c} \sigma_{\mu\alpha} P^\alpha \gamma_5 b, \quad (2)$$

where P is the momentum of the initial axial-vector state. The physical currents are written as

$$J_{1\mu} = \sin \theta J_\mu^A + \cos \theta J_\mu^B, \quad (3)$$

$$J_{2\mu} = \cos \theta J_\mu^A - \sin \theta J_\mu^B. \quad (4)$$

The lower state $B_{c1}(6743)$ is coupled to $J_{1\mu}$, while the upper state $B_{c1}(6750)$ is coupled to $J_{2\mu}$. The two-point sum-rule diagonalization gives $\theta \simeq 43.3^\circ$ in the numerical window used below.

The decay constants are defined by

$$\langle 0 | \bar{c} i \gamma_5 b | B_c(p) \rangle = \frac{m_{B_c}^2 f_{B_c}}{m_b + m_c}, \quad (5)$$

$$\langle 0 | \bar{c} \gamma_\rho b | B_c^*(p, \xi) \rangle = m_{B_c^*} f_{B_c^*} \xi_\rho, \quad (6)$$

$$\langle 0 | J_{i\mu} | B_{c1}^{(i)}(P, \eta) \rangle = f_i m_i \eta_\mu, \quad i = 1, 2. \quad (7)$$

For the pseudoscalar final state we parametrize the radiative matrix element as

$$\langle B_c(p) \gamma(q, \varepsilon) | B_{c1}^{(i)}(P, \eta) \rangle = e g_i [(\eta \cdot \varepsilon^*)(p \cdot q) - (\eta \cdot q)(p \cdot \varepsilon^*)]. \quad (8)$$

The corresponding width is

$$\Gamma(1^+ \rightarrow 0^- \gamma) = \frac{\alpha_{\text{em}}}{3} g_i^2 \left(\frac{m_i^2 - m_{B_c}^2}{2m_i} \right)^3. \quad (9)$$

For the vector final state we use the gauge-invariant tensor

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho\alpha\beta} p^\alpha q^\beta - (p \cdot q) \epsilon_{\mu\nu\rho\alpha} p^\alpha, \quad (10)$$

and define

$$\langle B_c^*(p, \xi) \gamma(q, \varepsilon) | B_{c1}^{(i)}(P, \eta) \rangle = e g_i^* \eta^\mu \varepsilon^{*\nu} \xi^{*\rho} V_{\mu\nu\rho}. \quad (11)$$

The width is evaluated by an explicit polarization sum.

III. QCD SUM RULES

For the $B_c \gamma$ channel the correlation function is

$$\Pi_{i\mu}(p, q) = i \int d^4x e^{ip \cdot x} \langle \gamma(q, \varepsilon) | T \{ J_{i\mu}(x) \bar{b}(0) i \gamma_5 c(0) \} | 0 \rangle. \quad (12)$$

For the $B_c^* \gamma$ channel the pseudoscalar current is replaced by $\bar{b} \gamma_\rho c$. Since both quarks are heavy, the leading OPE is obtained by attaching the photon perturbatively to either the charm line or the anti-bottom line. In the charm-emission topology the three denominators are

$$D_1 = (k + q)^2 - m_c^2, \quad D_2 = k^2 - m_c^2, \quad D_3 = (k - p)^2 - m_b^2, \quad (13)$$

while in the anti-bottom-emission topology they are

$$D_1 = k^2 - m_c^2, \quad D_2 = (k - p)^2 - m_b^2, \quad D_3 = (k - p + q)^2 - m_b^2. \quad (14)$$

The projected perturbative traces generate invariant amplitudes $\hat{\Pi}_A$ and $\hat{\Pi}_B$, which are then rotated into the physical basis,

$$\hat{\Pi}_1 = \sin \theta \hat{\Pi}_A + \cos \theta \hat{\Pi}_B, \quad (15)$$

$$\hat{\Pi}_2 = \cos \theta \hat{\Pi}_A - \sin \theta \hat{\Pi}_B. \quad (16)$$

The resulting form factors have the schematic form

$$g_1 = \frac{\exp[(m_1^2 + m_f^2)/(2M^2)]}{m_1 f_1 \mathcal{N}_f} \left(\sin \theta \hat{\Pi}_A + \cos \theta \hat{\Pi}_B \right), \quad (17)$$

$$g_2 = \frac{\exp[(m_2^2 + m_f^2)/(2M^2)]}{m_2 f_2 \mathcal{N}_f} \left(\cos \theta \hat{\Pi}_A - \sin \theta \hat{\Pi}_B \right), \quad (18)$$

where $\mathcal{N}_f = m_{B_c}^2 f_{B_c} / (m_b + m_c)$ for $B_c \gamma$ and $\mathcal{N}_f = m_{B_c^*} f_{B_c^*}$ for $B_c^* \gamma$.

The dimension-four gluon-condensate workbench uses the background-field heavy propagator expansion. The projected G^2 numerators were reduced in powers of D_1, D_2, D_3 . For the vector channel this gives 1095 reduced terms: 538 with all effective denominator powers positive and 557 contact/derivative-sector terms. A refined double-Borel support audit shows that the contact/derivative rows do not contain ordinary two-channel double poles. The apparent 131 active rows all depend on the crossed combination $(p-q)^2 = 2p^2 - P^2$, and are therefore kept as audit candidates rather than added to the physical double-Borel sum. Until the all-positive G^2 sector is integrated explicitly, we quote a conservative systematic envelope obtained from the same-window two-point G^2/pert ratios.

IV. NUMERICAL ANALYSIS

We use $m_b = 4.18$ GeV, $m_c = 1.27$ GeV, $m_{B_c} = 6.2749$ GeV, and $m_{B_c^*} = 6.3359$ GeV. The axial-state masses are $m_1 = 6.743$ GeV and $m_2 = 6.751$ GeV. The working Borel and continuum windows are

$$M^2 = 7, 8, 9 \text{ GeV}^2, \quad s_0 = 53, 54, 55 \text{ GeV}^2,$$

and the vector-current two-point sum rule for $f_{B_c^*}$ uses

$$M_V^2 = 5, 6, 7, 8 \text{ GeV}^2, \quad s_{0,V} = 40, 42, 44 \text{ GeV}^2.$$

The two-point residue grid gives

$$\theta = 43.29^\circ, \quad f_1 = 0.264[0.259, 0.269] \text{ GeV}, \quad f_2 = 0.722[0.709, 0.738] \text{ GeV}.$$

For the vector final-state decay constant we find

$$f_{B_c^*} = 0.435[0.400, 0.467] \text{ GeV}$$

in the standard vector-invariant convention.

Table II compares our results with representative model predictions. LHCb has observed the $B_c^+ \gamma$ structure, but the individual partial radiative widths are not measured [1, 2]; hence the comparison is with theory rather than direct width measurements.

The small $B_{c1}(6743) \rightarrow B_c \gamma$ width and the comparatively large $B_{c1}(6750) \rightarrow B_c^* \gamma$ width reflect the strong dependence of the radiative amplitude on the physical-current rotation

TABLE I. Present hard-QCDSR results. The first uncertainty is the Monte Carlo input/Borel-window uncertainty. The second uncertainty, where shown, is the conservative G^2 screening envelope.

Channel	g [GeV $^{-1}$]	Γ [keV]
$B_{c1}(6743) \rightarrow B_c \gamma$	$+0.0316[0.0269, 0.0370]$	$0.223^{+0.084}_{-0.061} \pm 0.025_{G^2}$
$B_{c1}(6750) \rightarrow B_c \gamma$	$-0.184[-0.205, -0.167]$	$8.01^{+1.89}_{-1.41} \pm 0.91_{G^2}$
$B_{c1}(6743) \rightarrow B_c^* \gamma$	$-0.0785[-0.0843, -0.0736]$	$74.3^{+11.2}_{-9.0} \pm 11.1_{G^2 \text{ scr.}}$
$B_{c1}(6750) \rightarrow B_c^* \gamma$	$+0.168[0.156, 0.180]$	$360^{+52}_{-48} \pm 54_{G^2 \text{ scr.}}$

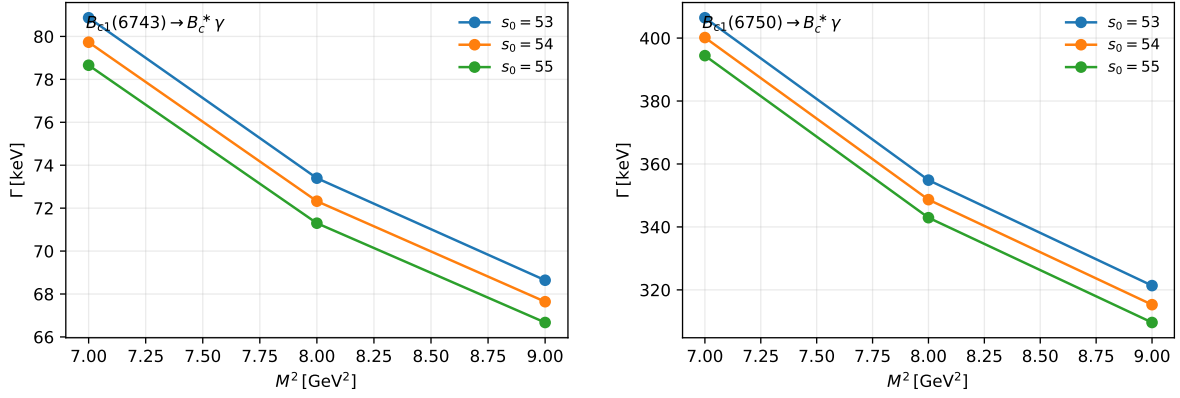


FIG. 1. Representative Borel stability of the vector final-state widths in the standard $f_{B_c^*}$ normalization.

and on the final-state spin. The vector channels are also sensitive to the normalization of $f_{B_c^*}$; using the reference value $f_{B_c^*} = 0.384 \pm 0.038$ GeV instead of the self-contained standard sum-rule value gives $\Gamma(6743 \rightarrow B_c^* \gamma) = 92.6[76.1, 117]$ keV and $\Gamma(6750 \rightarrow B_c^* \gamma) = 447[365, 564]$ keV.

V. CONCLUSION

We have calculated the leading hard-photon QCD sum-rule baselines for $B_{c1} \rightarrow B_c \gamma$ and $B_{c1} \rightarrow B_c^* \gamma$ transitions. The calculation uses physical axial-current residues from the two-point B_c -mixing sum rules and a self-contained vector-current sum rule for $f_{B_c^*}$. The resulting widths show a strong channel hierarchy: $B_{c1}(6743) \rightarrow B_c \gamma$ is very small, $B_{c1}(6750) \rightarrow B_c \gamma$

TABLE II. Comparison with representative theoretical predictions. Widths are in keV.

Publication	$6743 \rightarrow B_c \gamma$	$6743 \rightarrow B_c^* \gamma$	$6750 \rightarrow B_c \gamma$	$6750 \rightarrow B_c^* \gamma$
Godfrey [3]	13	60	80	11
Ebert–Faustov–Galkin [4]	18.4	78.9	132	13.6
Fulcher [5]	32.4	75.6	127.8	26.2
Gershtein et al. [6]	11.6	77.8	131.1	8.1
T. Y. Li et al. [7]	16.6	49	66.6	10.5
Q. Li et al. [8]	35	70	74	40
X. J. Li et al. [9]	30.1	47.8	64	25.6
Eichten–Quigg [10]	9.9	62.5	92.3	7.5
Bondar–Milstein [11]	22	75	47	2
This work	0.223	74.3	8.01	360

is at the few-keV level, and the vector final-state widths are much larger in the standard normalization.

The dimension-four gluon-condensate sector has been reduced algebraically and audited. In the vector channel the contact/derivative rows do not have ordinary two-channel double-pole support; they are crossed-contact audit candidates. The numerical results are therefore quoted as leading perturbative hard-QCDSR predictions with a conservative G^2 screening uncertainty. A future refinement can integrate the all-positive radiative G^2 terms explicitly, but the present calculation already gives a transparent QCDSR baseline for comparison with model predictions and future measurements of $B_c(1P)$ radiative decays.

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